Whole-Brain Geometric Representations for Discovering Structural Biomarkers of Neurodegenerative Disease in an Asia-Pacific Cohort

Thomas B Shaw[1,2,3], Akshit Beniwal[1], Robert D Henderson[3,4], Matthew C Kiernan[5,6], Shuangwu Liu[7], Muddasu S Keerthipriya[8], Atchayaram Nalini[8], Markus Barth[1,2], Sicong Tu[5,9*], Fernanda L Ribeiro[1,10*]

[1] School of Electrical Engineering and Computer Science, The University of Queensland, St Lucia, QLD, Australia, [2] Centre for Advanced Imaging, AIBN, The University of Queensland, St Lucia, QLD, Australia, [3] Department of Neurology, Royal Brisbane and Women's Hospital, Herston, QLD, Australia, [4]UQ Centre for Clinical Research, Herston Queensland, Australia, [5]Brain and Mind Centre, The University of Sydney, Sydney, NSW, Australia, [6]Neuroscience Research Australia, Randwick, NSW, Australia, [7] School of Nursing and Rehabilitation, Cheeloo College of Medicine, Shandong University, China, [8] Departments of Neurology and Neuroimaging and Interventional Radiology, National Institute of Mental Health and Neurosciences (NIMHANS), Bengaluru, Karnataka, India, [9] South Eastern Sydney Local Health District, Randwick, NSW, Australia, [10] Justus-Liebig University Giessen, * Senior Authors.

Introduction

Neurodegenerative diseases are characterised by spatially distributed brain changes that extend beyond canonical regions of interest and are difficult to capture with conventional voxel-based or region-wise summary measures. We require methods that respect the intrinsic geometry and topology of cortical structures. Geometric Deep Learning (GDL) generalises deep learning to non-Euclidean domains such as meshes and graphs, allowing models to operate directly on anatomically meaningful brain representations[1]. In this work, we apply GDL and vertex-wise analyses[2] to whole-brain structural MRI to identify distributed, disease-relevant patterns of brain morphology. Using Amyotrophic Lateral Sclerosis (ALS) as a test case for neurodegenerative disease, we evaluate whether geometric representations across the entire brain can reveal combinations of regions that optimally discriminate disease from health in a new, collaborative dataset collected across Australia, China, and India.

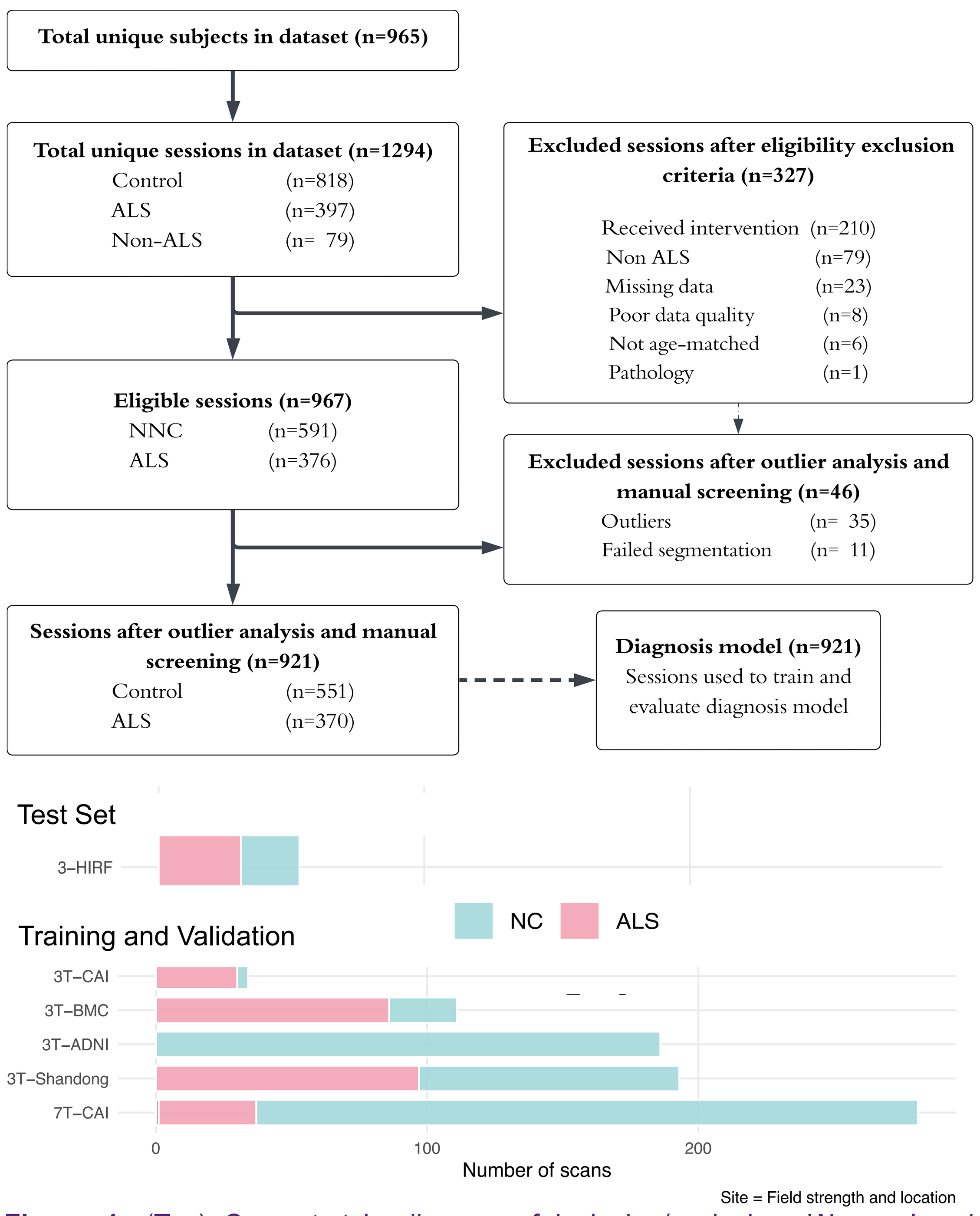

Figure 1: (Top) Consort-style diagram of inclusion/exclusion. We analysed 1,294 structural T1-weighted MRI scans comprising 397 scans from individuals with a clinically confirmed diagnosis of ALS[3], 783 age-matched healthy control scans (non-neurodegenerative controls – NC). Data were aggregated across multiple imaging sites associated with the Asia-Pacific MND Imaging Initiative, a collaborative initiative for biomarker discovery in ALS, which includes Australian, Chinese, and Indian sites and participants. (Bottom) dataset site breakdowns.

Methods

See Figure 1 for cohort details, Figure 2 for network details, and Figure 3 for methodology details.

Model performance was evaluated using k-fold cross-validation in an in-sample set, and in an independent unseen test set from the Royal Brisbane and Women's Hospital. Classification success was quantified using accuracy, recall (sensitivity), and F1 score, defined as the harmonic mean of precision and recall, to account for class imbalance and provide a balanced measure of model performance.

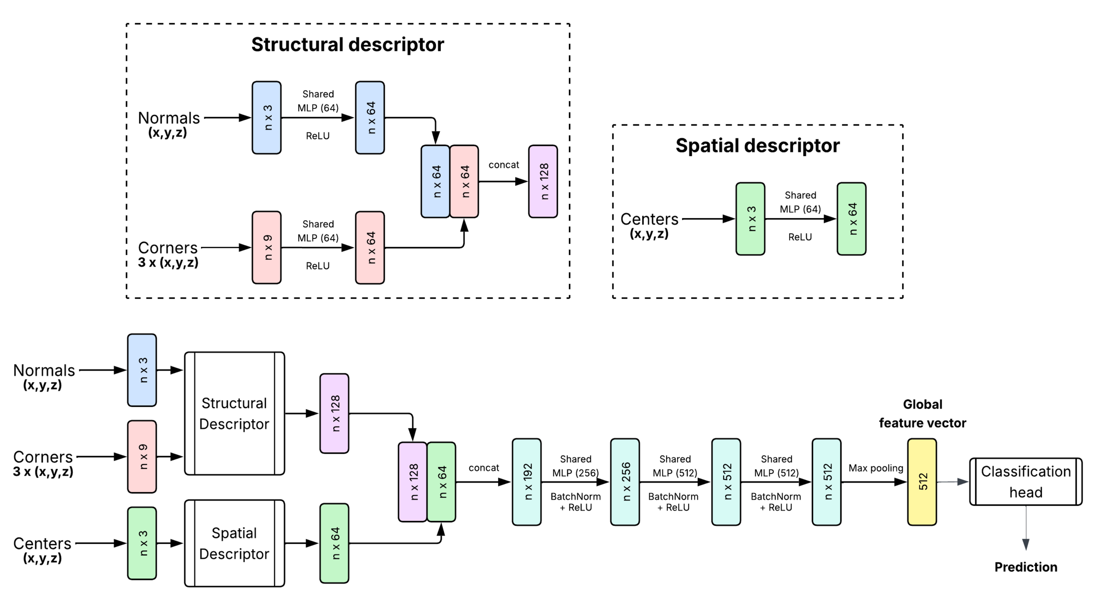

Figure 2: A modified MeshNet-style[5] geometric classifier applied directly to individual cortical surface meshes. Each triangular face carries a structural descriptor: its unit normal, encoding local orientation/cortical folding, and spatial descriptors (the face centroid plus its three corner-vertex coordinates, encoding position and shape). An EdgeConv backbone aggregates features across each face's three edge-adjacent neighbours over stacked layers, followed by global max-pooling to a per-region embedding and a fully-connected head predicting NC vs. ALS.

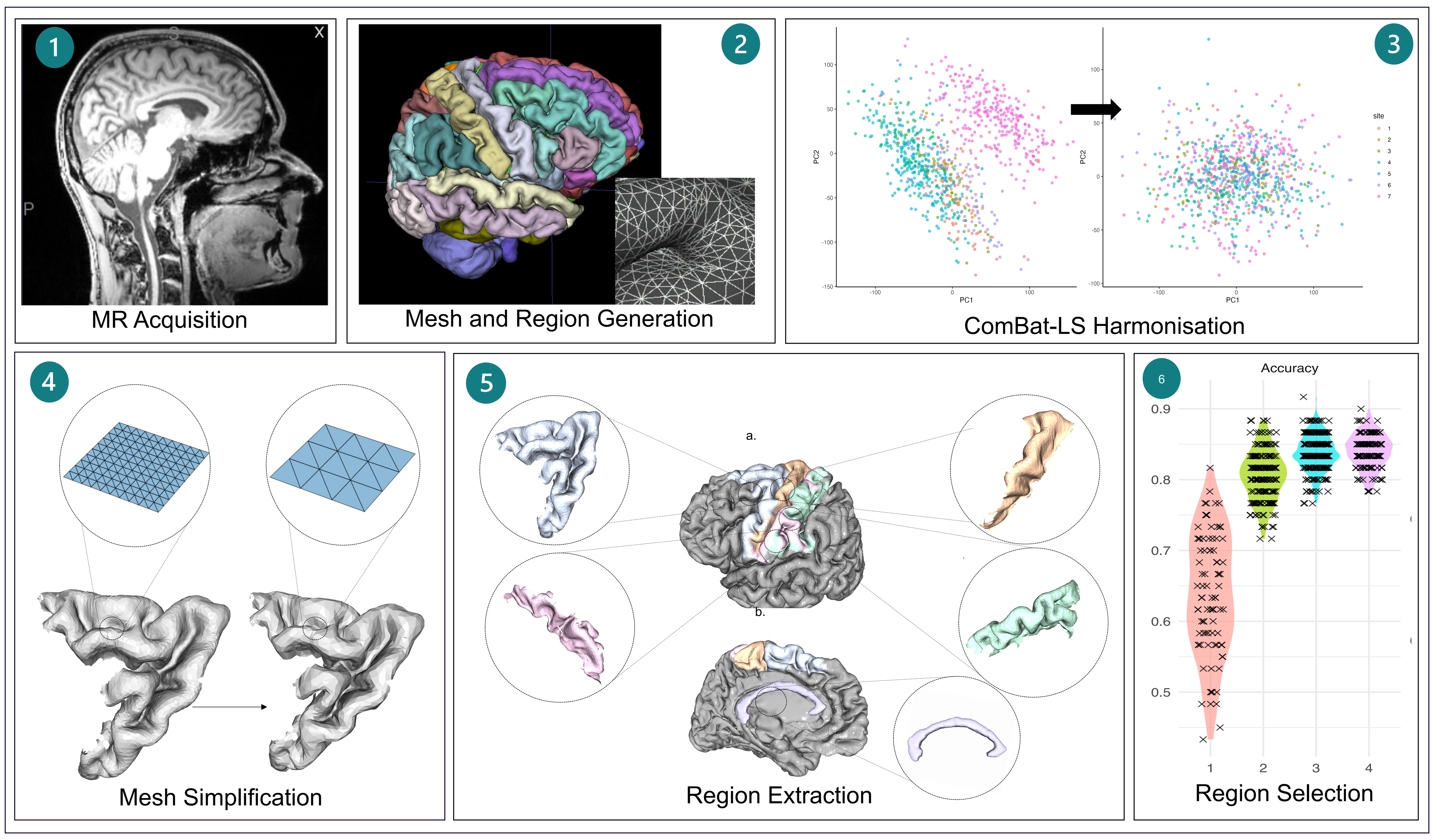

Figure 3: T1w images were acquired (1), then cortical structures were parcellated using the DKT atlas (2). Vertex-wise cortical thickness, sulcal depth, and curvature were extracted using FastSurfer, and region-wise summaries of these geometric measures were harmonised across sites using ComBat-LS[4] (3). We next analysed these metrics using VertexWiseR[3], with group differences assessed using Random Field Theory to control family-wise error. We modelled control (NC) vs ALS while accounting for both sex and age in the model (3). For each region, mesh-based geometric representations were extracted to capture shape and topological properties. Meshes were simplified (4) for computational efficiency and normalised to the proportional size of each ROI (5). All regions were first evaluated individually (single-region models), and regions were ranked by F1 score. Top-performing regions were retained and progressively combined, and a marginal gain analysis was performed to quantify the contribution of each region to predictive performance across multi-region combinations. This procedure guided systematic evaluation of 2, 3-, and 4-region models to identify subsets of regions whose joint geometry maximised predictive performance (6).

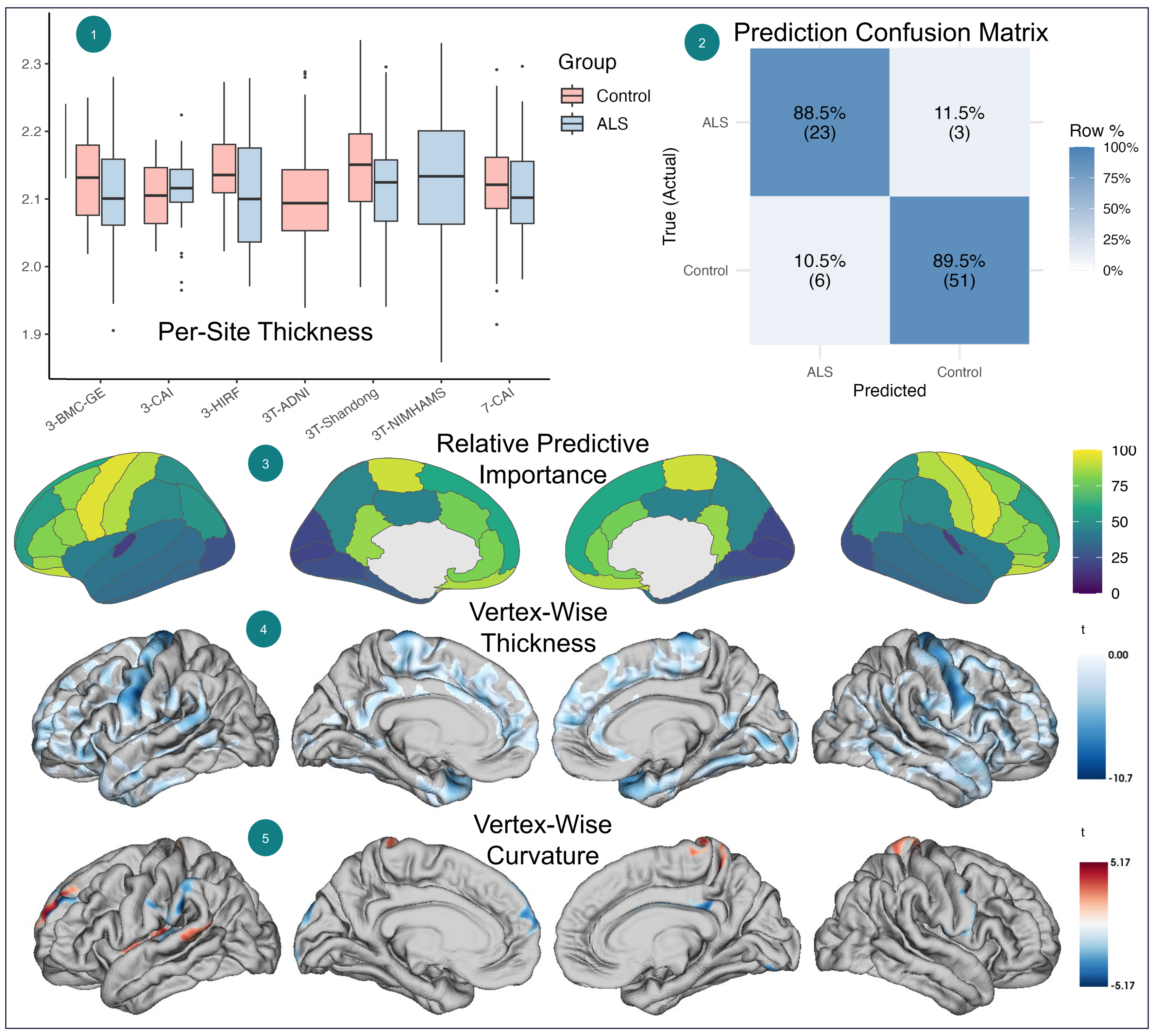

Figure 4: Across tested combinations, regions with the highest predictive importance included the pre-/paracentral gyrus, primary somatosensory cortex, caudal middle frontal gyrus, pars orbitalis, lateral orbitofrontal cortex, and isthmus of the cingulate. These regions contributed consistently to classification performance when aggregated across models. Regions contributing strongly to model performance overlapped with regions showing significant vertex-wise differences between ALS and NCs in cortical thickness, sulcal depth, or curvature (4,5). This correspondence suggests that the model's predictive signal was driven by localised geometric differences captured at the vertex level, which are diluted or obscured when reduced to region-wise summary measures.

Results

The final model achieved a cross-validated accuracy of 86% and an accuracy of 67.3% on independent test data, with corresponding recalls of 85% and 76.6% (Figure 4).

Performance improved when multiple regions were modelled jointly (Figure 4), indicating that neurodegenerative disease (ALS) is encoded in distributed geometric patterns rather than isolated regional abnormalities. A data-driven search over regional combinations identified an optimal subset of four anatomically distinct regions whose combined geometry yielded the strongest F1 scores, reflecting a favourable balance between precision and recall (Figure 4).

References

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Conclusion

This study demonstrates that whole-brain geometric representations can uncover distributed structural signatures of neurodegenerative disease in a data-driven yet neuroanatomically grounded manner. By operating directly on anatomically defined brain manifolds and integrating complementary geometric properties, GDL enables discovery of informative combinations of regions without strong a priori assumptions about disease localisation. The convergence between model-selected regions, their relative predictive importance, and independent vertex-wise measures of cortical thickness, sulcal depth, and curvature supports the biological plausibility of the learned representations. This framework is broadly applicable to other neurodegenerative conditions and supports a shift toward geometry-aware, whole-brain biomarkers for diagnosis, stratification, and clinical trial design.

t.shaw@uq.edu.au
Thomas B Shaw
https://about.uq.edu.au/experts/30967
https://thomshaw92.github.io/
School of Electrical Engineering and Computer Science